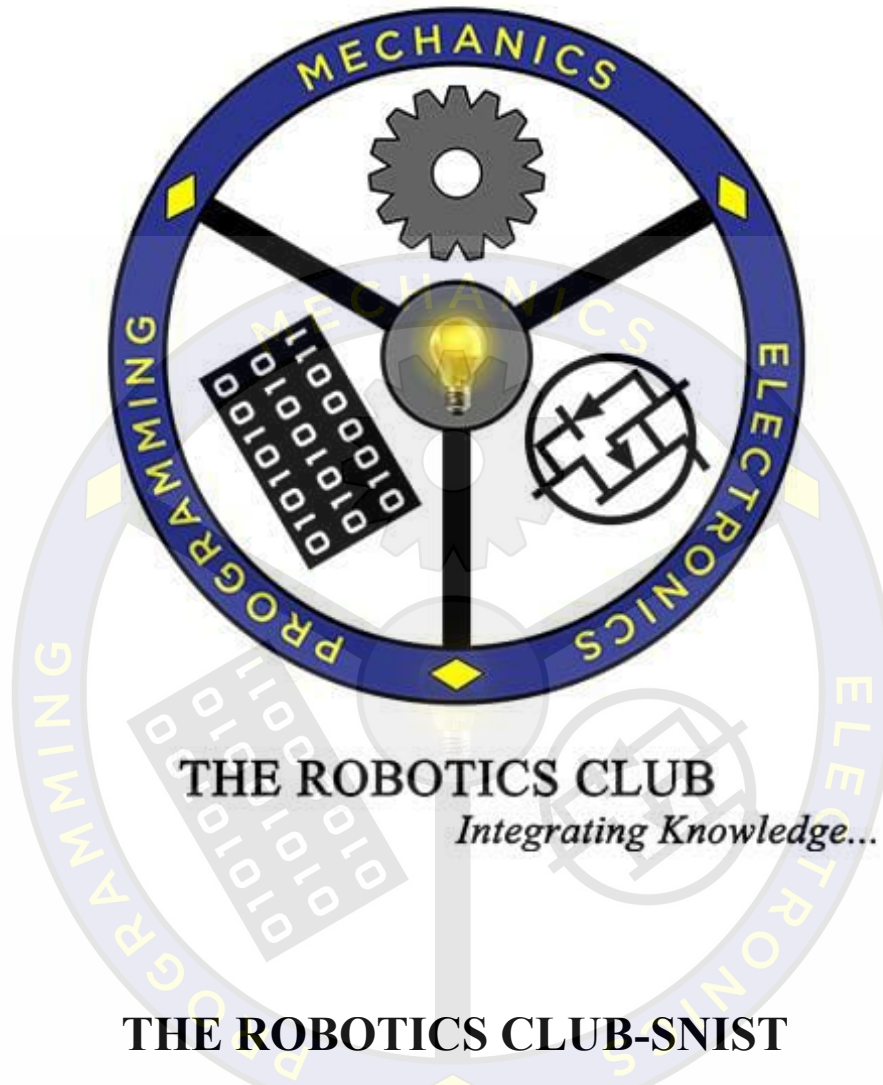


Project Report on
AUTOMATIC ROOFING SYSTEM



THE ROBOTICS CLUB-SNIST
SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
(AUTONOMOUS)

(Affiliated to JNTU University, Hyderabad)
Yamnapet, Ghatkesar, Hyderabad – 501301.

2024

CERTIFICATE

This is the project work titled '**AUTOMATIC ROOFING SYSTEM**' by 'Jayanth, Dhakshinya, Selvaraj Das, Mohan sai, Jeevana jyothi, Venkateswara rao, Ashritha, Anjali, Lokesh' under the mentorship of '**Tejaswi , Thanush**' and is a record of the project work carried out by them during the year 2023-2024 as part of INDUCTION under the guidance and supervision of

Mr. G. Kovindh Addhish
&
Mr. Aarushraj Puduchery
Joint Secretary for Technical Affairs

Mr. N V V S Narayana
The President of
THE ROBOTICS CLUB

Dr. A. PURUSHOTHAM

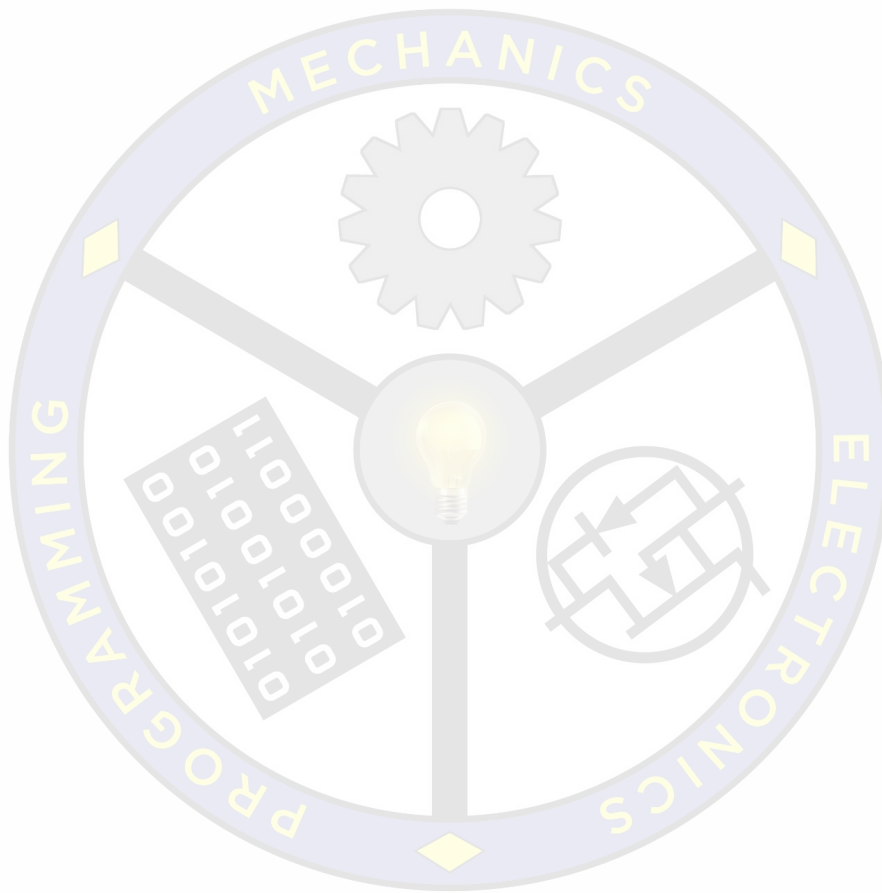
Faculty Advisor
Mechanical Department

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DECLARATION

The project work reported in the present thesis titled “**AUTOMATIC ROOFING SYSTEM**” is a record work done by Team 12 in **THE ROBOTICS CLUB** as a part of **INDUCTION-2024**.

No part of the thesis is copied from books/ journals/ Internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the project work done entirely by TEAM 06 and not copied from any other source.



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ACKNOWLEDGMENT

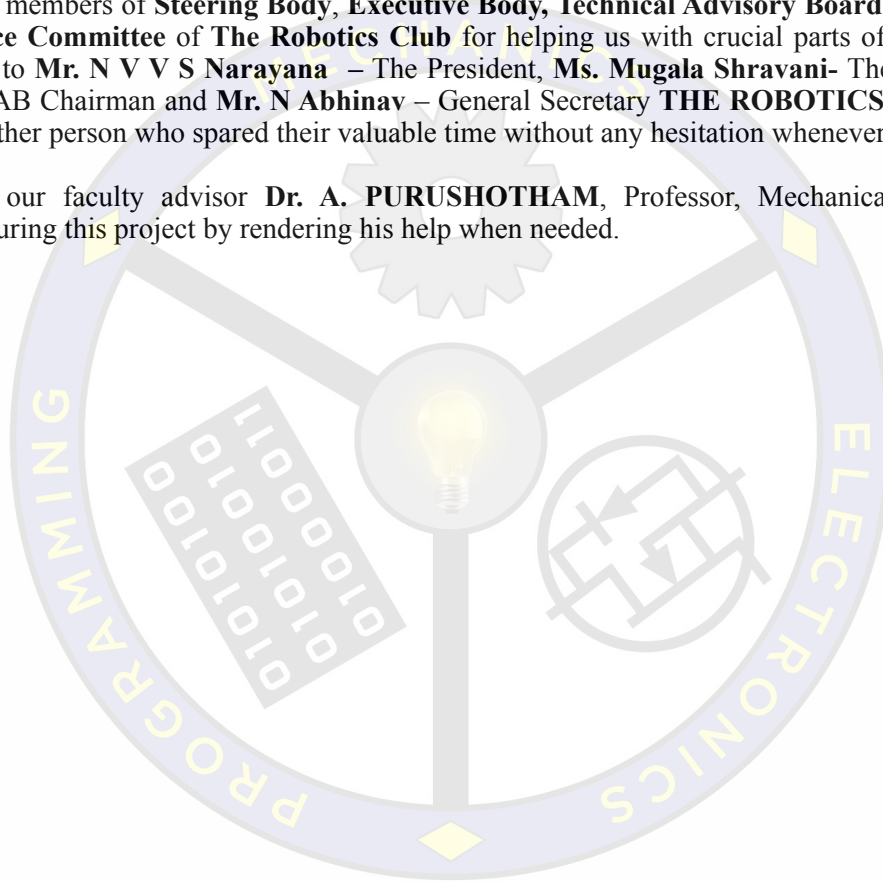
This project report is the outcome of the efforts of many people who have driven our passion to explore into implementation of **AUTOMATIC ROOFING SYSTEM**. We have received great guidance, encouragement and support from them and have learned a lot because of their willingness to share their knowledge and experience.

Primarily, we would like to express our gratitude to our mentors, **Thanush and Tejaswi**. Their guidance has been of immense help in surmounting various hurdles along the of our goal.

We thank our technical heads “**Mr.G. Kovidh Addhish & Mr. Aarushraj Puducherry**” for being with us till the end of the project completion.

We thank all the members of **Steering Body, Executive Body, Technical Advisory Board, Club’s Incubation and Competence Committee** of **The Robotics Club** for helping us with crucial parts of the project. We are deeply indebted to **Mr. N V V S Narayana** – The President, **Ms. Mugala Shravani**- The Vice President and **Ms Maliha** – SAB Chairman and **Mr. N Abhinav** – General Secretary **THE ROBOTICS CLUB** respectively and also every other person who spared their valuable time without any hesitation whenever we wanted.

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ABSTRACT
INDUCTION'23
TEAM-12
AUTOMATIC ROOFING SYSTEM

THE PROBLEM:

Agriculture is a critical sector, contributing approximately 70% of the country's revenue. However, farmers face significant challenges during heavy rainfall, which can wash away or destroy cultivated crops. This destruction leads to reduced plant growth and yields. Existing measures to mitigate these environmental hazards are often performed manually and are not always effective. Thus, there is a pressing need for an automated solution that can protect crops from heavy rainfall and conserve rainwater for various uses.

TEAM'S APPROACH TO SOLVE THE PROBLEM:

To address the challenges faced by farmers due to heavy rainfall, our team proposes an innovative solution utilizing Arduino UNO, a soil moisture sensor, a rain sensor, Li-ion batteries, BO motors with wheels, pulleys and polythene sheets. The system will be designed to automatically detect heavy rainfall and respond by deploying protective polythene sheets over the crops. The rain sensor will detect precipitation, triggering the Arduino Uno to activate the motors and pulleys that will extend the polythene sheets over the crops. Simultaneously, the soil moisture sensor will monitor soil saturation levels to ensure the deployment is necessary, preventing unnecessary activation during light rains. The Li-ion batteries will power the system, ensuring it operates independently of external power sources. This automated approach not only protects crops from damaging effects of heavy rainfall but also conserves rainwater, which can be redirected for irrigation or other agricultural uses. By automating the response to environmental hazards, this solution aims to improve crop yields, reduce labor for farmers, and enhance overall agricultural productivity.

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Automatic Roofing System

¹Venkateswara rao, ²Mohan Sai, ³Dhakshinya, ³Selvaraj Das, ⁴Jeevana Jyothi,
⁵Jayanth, ⁶Lokesh, ⁷Ashritha, ⁸Anjali
Sreenidhi Institute of Science and Technology
Hyderabad, Telangana, India

June 22, 2024

Abstract

As we know about 70 percentage of our country's revenue comes from agriculture. But during heavy rainfalls, the farmers face a lot of problems because their cultivated crops get washed off or destroyed. Which reduces the plant growth in turn reduces yield. As human beings, we cannot control the natural phenomenon such as rain, moisture, etc. Some of the measures are taken against this environmental hazard but they are performed manually. Here comes the need for automation. So to avoid this problem this project is designed, which helps in protecting the crops from heavy rainfall and saving that rainwater to use for other purposes. In this system, an automatic roof is used which works by taking the signals from the rain sensor and soil moisture sensors and covers the whole field to protect it from heavy rains. Whenever there is rainfall the rain sensor gets activated. The soil moisture sensor is used to check the water content in the soil. Whenever there is rain, the rain sensor gets on and the Sensor output value below a threshold indicates heavy rain, while the sensor output value above the threshold value indicates light rain or no rain. If is a heavy rain fall then the BO motor starts which opens the roof automatically to close the field using a polythene sheet. Here we will be using a Li-ion batteries for power supply. The roof will be kept on a slope so that the water can be collected in a container for feeding animals, washing, cooking, etc., and can also be reused to sprinkle it back.

1 INTRODUCTION

Heavy rain can cause a lot of problems for farmers. Downpours can flood fields, erode topsoil, and damage delicate crops, leading to lost harvests and income.

This project aims to develop an Automatic Roofing System designed to shield crops from heavy rainfall and save rainwater for reuse. The system utilizes a combination of rain sensors, soil moisture sensors, and respond accordingly. When rain is detected, the rain sensor activates the system.

The soil moisture sensor assesses the water content in the soil, and the rain sensor helps predict rainfall. Upon detecting heavy rain, the system triggers a BO motor to deploy a polythene sheet, covering the field and protecting the crops. The system is powered by Li-ion batteries, ensuring it operates using renewable energy. The roof is inclined to allow collected rainwater to flow into a container, which can be used for animal feeding, washing, cooking, and irrigation, promoting sustainable water management.

This system has the potential to be a valuable asset to farmers by safeguarding crops during downpours and reducing the need for manual work.

2 LITERATURE SURVEY

A review of existing literature reveals a diverse array of methods for crop protection and water conservation, emphasizing their critical roles in sustainable agriculture. Historically, farmers have relied on a combination of traditional and modern techniques to shield crops from adverse weather conditions and to conserve water, which is vital for crop growth. Early approaches primarily involved manual interventions, such as covering crops with tarps or manually adjusting irrigation schedules based on observed soil moisture levels. While these techniques were somewhat effective, they were labor-intensive and often inefficient.

With technological advancements, semi-automated systems emerged. These systems often utilize sensors to monitor rainfall and soil moisture but still require human intervention to execute protective measures. For instance, a semi-automated system might include an alert mechanism that notifies a farmer to cover crops or adjust irrigation when certain environmental thresholds are reached. Despite the progress made with these semi-automated solutions, there remains a significant gap in the literature concerning fully automated systems that seamlessly integrate various components, including rain detection, soil moisture analysis, and renewable energy sources.

Existing research highlights the use of various sensors to detect rainfall, ranging from simple rain gauges to sophisticated meteorological devices. These sensors can trigger protective measures, such as deploying covers over crops or modifying irrigation schedules. Accurate soil moisture

analysis is also crucial for effective water conservation. Different types of soil moisture sensors, including tensiometers and dielectric sensors, have been studied to ensure precise moisture levels. Integrating these sensors with automated irrigation systems helps to provide crops with optimal water, reducing wastage and preventing water stress.

Furthermore, incorporating renewable energy sources, such as solar panels or wind turbines, into agricultural practices can enhance sustainability. Literature suggests that renewable energy can effectively power automated systems, making them more eco-friendly and reducing reliance on non-renewable energy sources. Despite these advancements, comprehensive solutions that integrate rain detection, soil moisture analysis, and renewable energy into a fully automated system are scarce. This project aims to address this gap by combining these elements into a holistic solution, thereby advancing the field of agricultural technology and promoting more sustainable farming practices.

3 SYSTEM ARCHITECTURE

3.1 Hardware

3.1.1 Arduino UNO

The Arduino Uno is a versatile microcontroller board widely favored in robotics for its ease of use and robust capabilities. It features an ATmega328P microcontroller at its core, offering 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, and a 16 MHz quartz crystal. These specifications make it ideal for interfacing with sensors, motors, and other electronic components essential in robotics. The Uno's programmability via the Arduino IDE simplifies code development, allowing users to write and upload programs quickly. Its compact size and compatibility with various shields and modules further enhance its utility in creating sophisticated robotic systems. Overall, the Arduino Uno's affordability, accessibility, and rich ecosystem of community support and resources make it an indispensable tool for robotics enthusiasts and professionals alike.

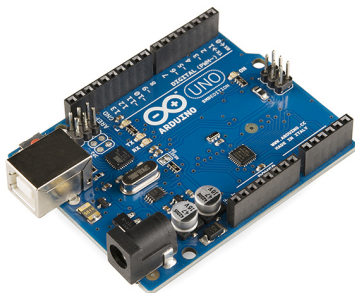


Figure 1: Arduino UNO

3.1.2 Bread board

An electronics breadboard (as opposed to the type on which sandwiches are made) is actually referring to a solder-less breadboard. These are great units for making temporary circuits and prototyping, and they require absolutely no soldering.

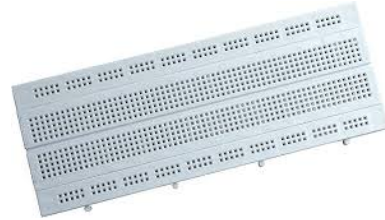


Figure 2: Bread board

3.1.3 BO motor

A BO motor is a small, lightweight DC gear motor commonly used in robotics projects. It's known for being affordable and easy to use. BO motors come in various versions with different speeds and features, but they typically offer high torque (turning force) for their size. This makes them suitable for powering small robots and other mechanisms that need to move steadily or carry a light load. They are also relatively quiet compared to some other motors. BO motors are a good choice for beginners in robotics due to their affordability, ease of use, and versatility in powering small creations.



Figure 3: BO motor

3.1.4 Motor driver

The L298N motor driver is a dual H-bridge motor driver integrated circuit (IC) used to control the direction and speed of two DC motors or a single stepper motor. It can handle a voltage range of 5V to 35V and provide a current up to 2A per channel. The L298N features built-in diodes for back EMF protection, making it robust for various applications. It includes input pins for controlling motor direction and enable pins for controlling motor speed through pulse-width modulation (PWM). Widely used in robotics and automation projects, the L298N offers a simple and efficient way to drive motors from microcontrollers like Arduino and Raspberry Pi.

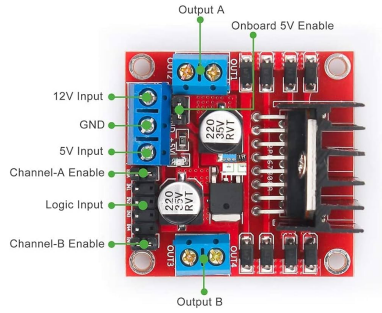


Figure 4: motor driver

3.1.5 Servo motor

A servo motor is a rotary actuator designed for precise control of angular position, velocity, and acceleration. It consists of a motor coupled with a sensor for position feedback, typically a potentiometer or encoder. An electronic circuit interprets signals from a control system and adjusts the motor's movement accordingly. Servo motors are widely used in robotics, automation, and remote-controlled devices due to their accuracy and reliability. They come in various sizes and power ratings, making them suitable for applications ranging from small hobby projects to industrial machinery.



Figure 5: servo motor

3.1.6 Li-ion batteries

Lithium-ion batteries (Li-ion) are rechargeable energy storage devices known for their high energy density and lightweight nature. They are widely used in portable electronics such as smartphones, laptops, and tablets, as well as in electric vehicles and renewable energy storage systems. They consist of several key components including a cathode (typically made of lithium metal oxide), an anode (commonly graphite), and an electrolyte that allows for the movement of lithium ions between the electrodes during charging and discharging cycles. Li-ion batteries operate by moving lithium ions between the positive and negative electrodes during charge and discharge cycles. Their development in the 1980s revolutionized portable electronics, offering longer-lasting and more efficient power solutions compared to earlier battery technologies. Despite their advantages, they require careful handling and monitoring to prevent safety issues such as overheating and battery degradation over time.



Figure 6: Li-ion battery

3.1.7 Rain sensor

A rain sensor is a device that detects rainfall. It typically consists of two main parts: A sensing plate that is exposed to the rain. This plate is often made of a material that conducts electricity when wet. A circuit board that processes the signal from the sensing plate. This circuit board may output a digital signal (indicating rain or no rain) or an analog signal (indicating the intensity of the rain). The rain sensor works by measuring the change in conductivity of the sensing plate when it gets wet. When rain falls on the plate, it becomes more conductive. The circuit board detects this change in conductivity and outputs a corresponding signal.



Figure 7: Rain sensor

3.1.8 Wheels

Wheels with good grip are used so that the bot moves freely through the pipes. They reduce friction, making it easier to transport heavy loads and navigate obstacles. We can traverse through rough terrain as they have good grip. The ones we are using are made of plastic and rubber.



Figure 8: Wheels



Figure 9: Soil moisture sensorsoil moisture sensor

3.1.9 Soil moisture sensor

A soil moisture sensor is an electronic device designed to measure the water content in soil. It typically consists of metal probes or sensors that are inserted into the soil to detect moisture levels. These probes are connected to electronic circuitry within the sensor, which processes the signals and converts them into a readable output. The output can be in the form of analog voltage, digital signal, or transmitted wirelessly via protocols like Bluetooth or Wi-Fi. Soil moisture sensors require power, usually from batteries or external sources, to operate the electronic components. They are often housed in durable enclosures to protect them from environmental conditions. These sensors are widely used in agriculture, environmental monitoring, and landscaping to optimize irrigation schedules and ensure efficient water use, thereby promoting healthier plant growth and conservation of water resources.

3.1.10 Jumper Wires

A jumper wire is an electrical wire or group of wires used to connect circuits without soldering. They have connectors or pins at their ends. Depending upon the configuration of end connectors, they are classified into three types: male-to-male, male-to-female and female-to-female.



Figure 10: Jumper Wires

3.2 Software

3.2.1 Arduino IDE

The Arduino Integrated Development Environment(IDE) is a software platform used for writing, compiling, and uploading code to Arduino microcontroller boards. It provides a user-friendly interface with features like a text editor for writing code, a message area, a text console, a tool bar with buttons for common functions, and a series of menus. The Arduino IDE supports the C and C++ programming languages using special rules of code structuring. It simplifies the process of programming microcontrollers, making it accessible for beginners and useful for advanced users in developing a wide range of electronic projects, from simple sensors to complex automation systems.

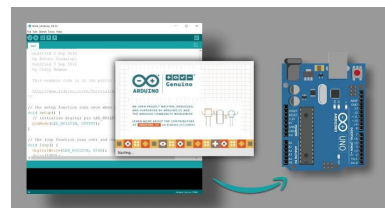


Figure 11: Arduino IDE

3.2.2 Fusion 360

Fusion 360 is a computer-aided designing (CAD) software application for 3-D modelling and simulation. It's other functions include computer-aided manufacturing (CAM) and computer-aided engineering (CAE), as well as designing printed circuit boards.

Fusion 360 provides powerful 3D modeling tools that allow users to create complex 3D models of products and components. It supports parametric modeling, direct modeling, and sculpting.

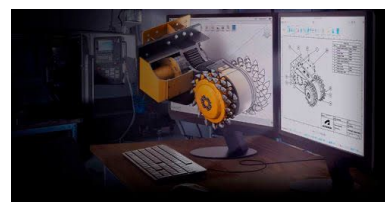


Figure 12: Fusion 360

3.2.3 Fritzing

Fritzing is an open-source electronic design automation (EDA) software to design electronics hardware, printed circuit boards and schematic circuit diagrams. It is an offshoot of the Processing programming language and the Arduino micro-controller.

Fritzing allows users to create electronic circuits by dragging and dropping various components (e.g., resistors, LED's, micro-controllers) onto a virtual breadboard. Users

can connect components by drawing wires to represent electrical connections.

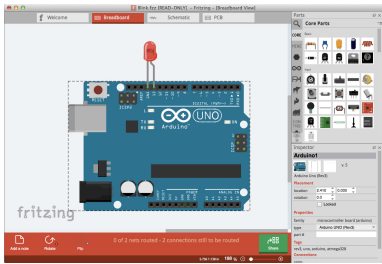


Figure 13: Fritzing

4 WORKING MECHANISM

The Automatic Roofing System addresses the critical issue of protecting crops during heavy rainfall while maximizing rainwater utilization in agriculture. The system operates using an automatic roof mechanism that integrates advanced sensors for real-time environmental monitoring. A rain sensor detects the onset and intensity of rainfall, triggering the system into action. Concurrently, soil moisture sensors assess soil water content, ensuring crops receive optimal irrigation levels.

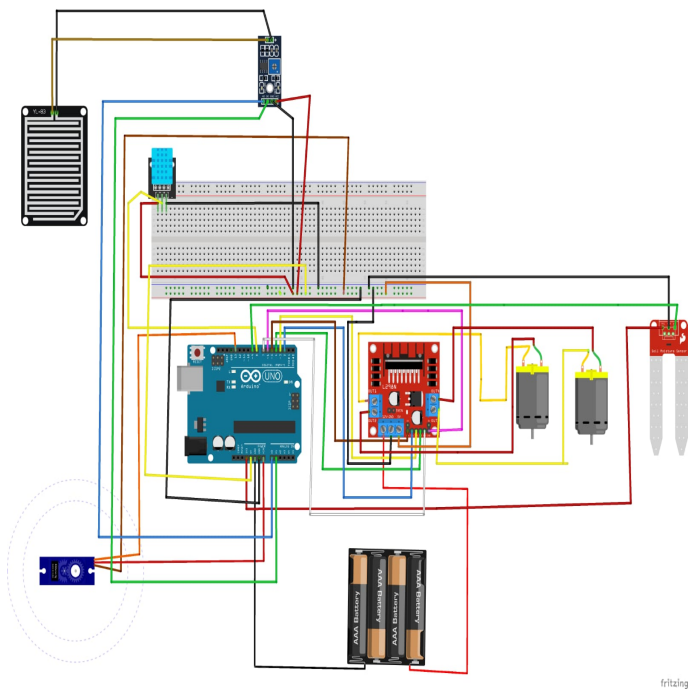


Figure 14: Circuit Diagram

Upon detecting heavy rain, indicated by the rain sensor output falling below a predefined threshold, a BO motor activates to automatically deploy a polythene sheet roof over the field. This protective cover shields crops from excessive rain, minimizing damage and promoting healthier plant growth. The roof is designed with a sloped surface to

efficiently channel rainwater into collection containers positioned strategically around the field. This collected rainwater serves multiple purposes, including feeding animals, domestic uses like washing and cooking, and potentially irrigation during dry spells, enhancing agricultural sustainability.

Powering this innovative system are Li-ion batteries, chosen for their reliability and energy efficiency. These batteries store energy harvested from solar panels, ensuring continuous operation without reliance on grid power. This renewable energy approach not only reduces operational costs but also aligns with sustainable farming practices. Regular maintenance and monitoring of sensors and battery health are essential to ensure optimal performance and longevity of the system, safeguarding crops and maximizing agricultural productivity year-round.

5 RESULTS

Upon the completion and deployment of the Automatic Roofing System, remarkable improvements in crop protection and water management were observed. One of the primary achievements of the project was the elimination of direct human intervention during heavy rainfall, significantly reducing crop damage and promoting healthier plant growth. The system, equipped with a responsive rain sensor, ensured timely deployment of the polythene sheet roof via a BO motor, effectively shielding crops from excessive rain. The integration of soil moisture sensors maintained optimal soil conditions, preventing over-saturation. Video streaming capabilities enabled real-time monitoring and assessment of the field conditions. The efficient design of the sloped roof allowed for the collection and repurposing of rainwater, supporting various agricultural and domestic uses. The system operated continuously and cost-effectively, powered by Li-ion batteries charged through batteries. Overall, the Automatic Roofing System was a successful solution that significantly enhanced agricultural productivity and sustainability.

a

6 FUTURE ENHANCEMENTS

Potential improvements to the system include integrating advanced weather prediction algorithms and machine learning models to improve the accuracy of rainfall detection and response times. Additionally, implementing a remote monitoring and control system via a mobile app would allow farmers to oversee and manage the system in real-time, regardless of their location. Upgrading the energy system to include hybrid power sources, such as wind turbines alongside solar panels, could ensure continuous operation even during low sunlight periods. Moreover, enhancing the material of the polythene sheets to a more durable and weather resistant alternative would increase the system's longevity and reliability. Expanding the system to include automated irrigation control based on soil moisture data could further

optimize water usage and crop health. Lastly, incorporating data analytics to track and predict crop growth patterns and environmental impacts would provide valuable insights for better farming practices and decision-making.

7 CONCLUSION

To conclude, this project introduces an Automatic Roofing System designed to protect agricultural crops from heavy rainfall and efficiently utilize rainwater. Using sensors like rain detectors, soil moisture sensors, the system automatically deploys a polythene sheet roof over the field upon detecting heavy rain, minimizing crop damage. Powered by Li-ion batteries charged by solar panels, it operates sustainably and reduces reliance on external power sources. This solution addresses the challenges posed by unpredictable weather, aiming to enhance crop yield and promote resource conservation in agriculture.

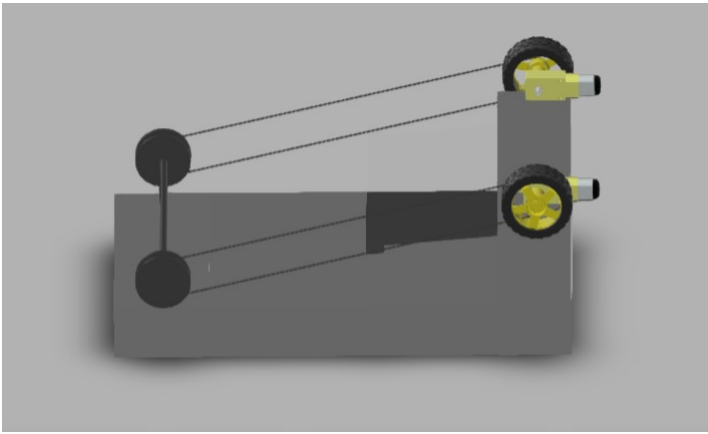


Figure 15: CAD Design of project

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- [3] University of Massachusetts Amherst - Retractable Roof Greenhouses and Shadehouses.
- [4] <https://www.researchgate.net/publication/323176442> Automatic Smart Roof System Using Arduino and Rain Drop Sensor
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- [6] Mechatronics: Principles and Applications Book by Godfrey C. Onwubolu
- [7] The Internet of Things: Key Applications and Protocols Book by David Boswarthick, Olivier Hersent, and Omar Elloumi

8 SOURCE CODE

```
#include <Servo.h>
// Define pin connections
int rainSensorPin = A0; // Rain sensor analog input
int moistureSensorPin = A1; // Moisture sensor analog input
int motorDriverPinA = 2; // Motor driver pin 1 (for A)
int motorDriverPinB = 4;
int motorDriverPinC = 7;
int motorDriverPinD = 8;
int enableAPin = 11;
int enableBPin = 3;
int servoPin = 10;
// Define Servo motor
Servo servo;

// Define threshold values for rain sensor
int heavyRainThreshold = 400;
int moderateRainThreshold = 700;

void setup() {
    Serial.begin(9600);
    pinMode(motorDriverPinA, OUTPUT);
    pinMode(motorDriverPinB, OUTPUT);
    pinMode(motorDriverPinC, OUTPUT);
    pinMode(motorDriverPinD, OUTPUT);
    pinMode(enableAPin, OUTPUT);
    pinMode(enableBPin, OUTPUT);
    pinMode(rainSensorPin, INPUT);
    pinMode(moistureSensorPin, INPUT);
    servo.attach(servoPin);
    servo.write(0);
    stopMotor(); // Stop the motor initially
}

void loop() {
    int rainSensorValue = analogRead(rainSensorPin);
    int moistureSensorValue = analogRead(moistureSensorPin);
    Serial.print("Rain sensor value: ");
    Serial.println(rainSensorValue);
    Serial.print("Moisture sensor value: ");
    Serial.println(moistureSensorValue);
    delay(2000);
    // Check rain sensor value and control motor
    if (rainSensorValue > heavyRainThreshold && moistureSensorValue < moderateRainThreshold) {
        // Heavy rain, cover the field
        Serial.println("Heavy rain, covering the field");
        moveMotorBackward();
        delay(2000);
        stopMotor();
        //delay(5000);
    }
    else if (heavyRainThreshold < rainSensorValue && moderateRainThreshold < rainSensorValue) {
        // Moderate rain, do nothing
        Serial.println("Moderate rain, no action");
        delay(2000);
        stopMotor();
    }
}
```

```

//delay(5000);
}
else
{
    // No rain, open the field
    Serial.println("No rain, opening the field");
    moveMotorForward();
    delay(2000);
    stopMotor();
}

// Check moisture sensor value and control servo motor

if (moistureSensorValue > 900)
{
    Serial.println("Soil is dry. Watering...");
    servo.write(180);
}
else
{
    Serial.println("Soil is wet. No water needed.");
    servo.write(0);
}
}

void moveMotorForward()
{
    Serial.println("field covered");
    digitalWrite(motorDriverPinA,HIGH);
    digitalWrite(motorDriverPinB,LOW);
    digitalWrite(motorDriverPinC,LOW);
    digitalWrite(motorDriverPinD,HIGH);
    analogWrite(enableAPin, 200); // Full speed forward (255 is maximum speed)
    analogWrite(enableBPin, 200);
    delay(2000); // Open roof for 5 seconds
}

void moveMotorBackward() {
    digitalWrite(motorDriverPinA,LOW);
    digitalWrite(motorDriverPinB,HIGH);
    digitalWrite(motorDriverPinC,HIGH);
    digitalWrite(motorDriverPinD,LOW);
    analogWrite(enableAPin, 200);
    analogWrite(enableBPin, 200);
    delay(2000); // Open roof for 5 seconds
}

void stopMotor() {
    digitalWrite(motorDriverPinA, LOW);
    digitalWrite(motorDriverPinB, LOW);
    digitalWrite(motorDriverPinC, LOW);
    digitalWrite(motorDriverPinD, LOW);
    analogWrite(enableAPin, 0);
    analogWrite(enableBPin, 0);
}

```

8.1 List of Expenses

S No.	Title	Cost
1	Arduino UNO	480
2	Soil Moisture sensor	250
3	Rain sensor	100
4	Li-ion batteries	400
5	Bread board	145
6	Jumper wires	120
7	Polythene sheet	50
8	BO Motors	50
9	Servo motor	110